

TF-blood_vs_primary

2026-04-22

Setup and data loading

```
library(tidyverse)
library(here)
library(corrplot)
data <- readxl::read_xlsx(here('data', 'processed', 'PCAWG_primary.xlsx'))
clean_data <- data
summary(clean_data)
```

donor_id	cancer_type	alt_status	tf_primary_rate
Length:2014	Length:2014	Length:2014	Min. : 0.00000
Class :character	Class :character	Class :character	1st Qu.: 0.00000
Mode :character	Mode :character	Mode :character	Median : 0.03931
			Mean : 0.13568
			3rd Qu.: 0.10251
			Max. :19.73776

tf_blood_rate	telomere_insertion_rate	telomere_content_log2
Min. :0.00000	Min. :0.00000	Min. : -4.00250
1st Qu.:0.00000	1st Qu.:0.00000	1st Qu.: -0.92876
Median :0.02628	Median :0.00000	Median : -0.52968
Mean :0.04660	Mean :0.03307	Mean : -0.43589
3rd Qu.:0.05967	3rd Qu.:0.02195	3rd Qu.: -0.07054
Max. :1.53097	Max. :5.38979	Max. : 3.84469

TERT_FPKM	ATAGGG_singleton_dist	CTAGGG_singleton_dist
Min. : 0.00000	Min. : -4.35757	Min. : -3.49674
1st Qu.: 0.02069	1st Qu.: -0.68156	1st Qu.: -0.38418

Median : 0.10630	Median : -0.03350	Median : 0.05423
Mean : 0.90171	Mean : -0.06042	Mean : 0.06588
3rd Qu.: 0.37979	3rd Qu.: 0.57645	3rd Qu.: 0.53662
Max. : 53.76790	Max. : 4.47470	Max. : 7.80772
NA's : 1166	NA's : 28	NA's : 2

GTAGGG_singleton_dist	TAAGGG_singleton_dist	TCAGGG_singleton_dist
Min. : -2.981384	Min. : -4.67195	Min. : -4.27208
1st Qu.: -0.359368	1st Qu.: -0.53849	1st Qu.: -0.56813
Median : 0.017444	Median : 0.08092	Median : 0.02901
Mean : 0.001582	Mean : 0.06257	Mean : 0.08148
3rd Qu.: 0.352901	3rd Qu.: 0.67706	3rd Qu.: 0.70046
Max. : 4.253404	Max. : 3.87943	Max. : 7.64722
	NA's : 28	NA's : 18

TGAGGG_singleton_dist	TTCGGG_singleton_dist	TTGGGG_singleton_dist
Min. : -5.27154	Min. : -3.848796	Min. : -3.109601
1st Qu.: -0.37005	1st Qu.: -0.568133	1st Qu.: -0.417046
Median : 0.06673	Median : 0.007297	Median : -0.019229
Mean : 0.07103	Mean : 0.102063	Mean : 0.002303
3rd Qu.: 0.46860	3rd Qu.: 0.573244	3rd Qu.: 0.429475
Max. : 6.67267	Max. : 8.739318	Max. : 4.584889
	NA's : 5	NA's : 2

TTTGGG_singleton_dist	Specimen.Type.Summary
Min. : -4.1941	Length: 2014
1st Qu.: -0.5610	Class : character
Median : -0.1030	Mode : character
Mean : -0.1309	
3rd Qu.: 0.3465	
Max. : 2.7484	

Distributions

Distributions - summary

```
summary(clean_data$tf_blood_rate)
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.00000	0.02628	0.04660	0.05967	1.53097

```
summary(clean_data$tf_primary_rate)
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.00000	0.03931	0.13568	0.10251	19.73776

```
zero_blood <- sum(clean_data$tf_blood_rate == 0)/nrow(clean_data) * 100  
zero_primary <- sum(clean_data$tf_primary_rate == 0)/nrow(clean_data) * 100  
  
cat("Percentuale di zeri in tf_blood_rate: ", zero_blood, "%\n")
```

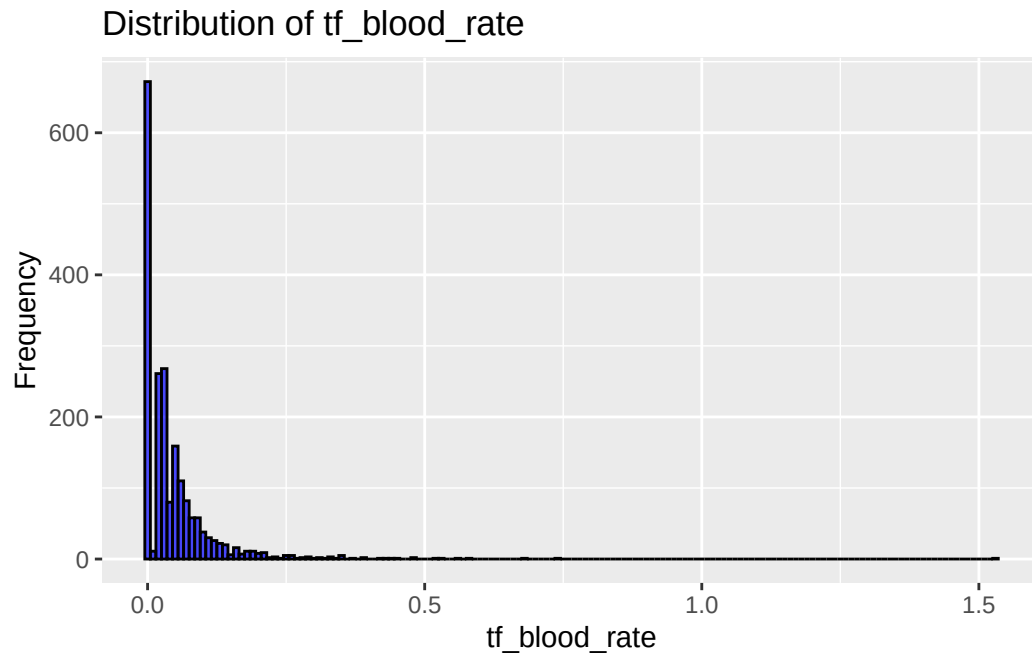
Percentuale di zeri in tf_blood_rate: 33.36643 %

```
cat("Percentuale di zeri in tf_primary_rate: ", zero_primary, "%\n")
```

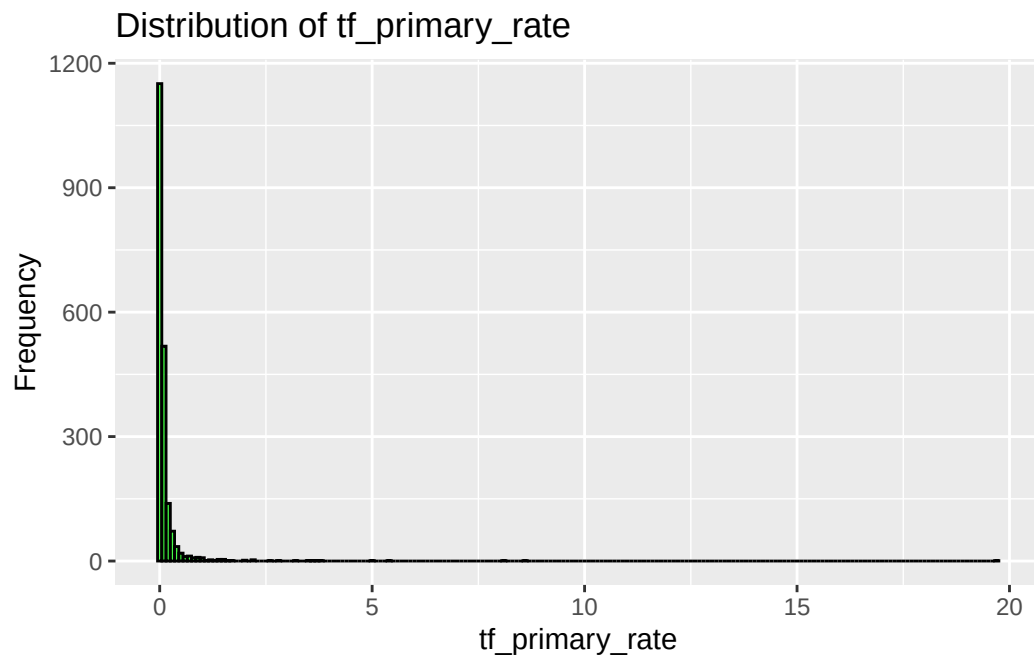
Percentuale di zeri in tf_primary_rate: 27.95432 %

Distributions - visual

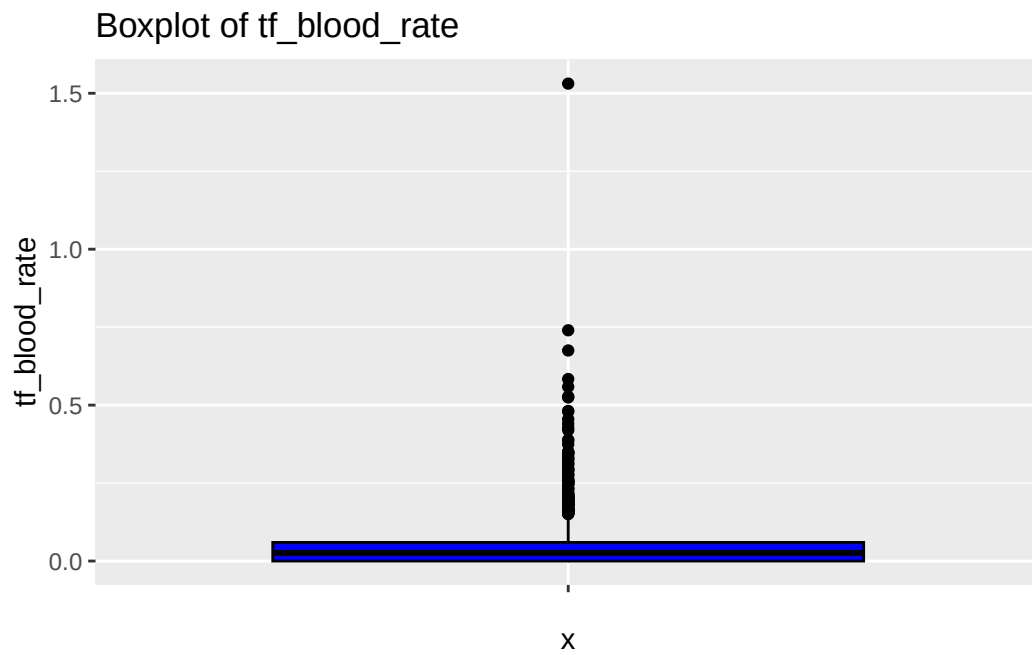
```
clean_data %>%  
  ggplot(aes(x = tf_blood_rate)) +  
  geom_histogram(binwidth = 0.01, fill = "blue", color = "black", alpha = 0.7) +  
  labs(title = "Distribution of tf_blood_rate", x = "tf_blood_rate", y = "Frequency")
```



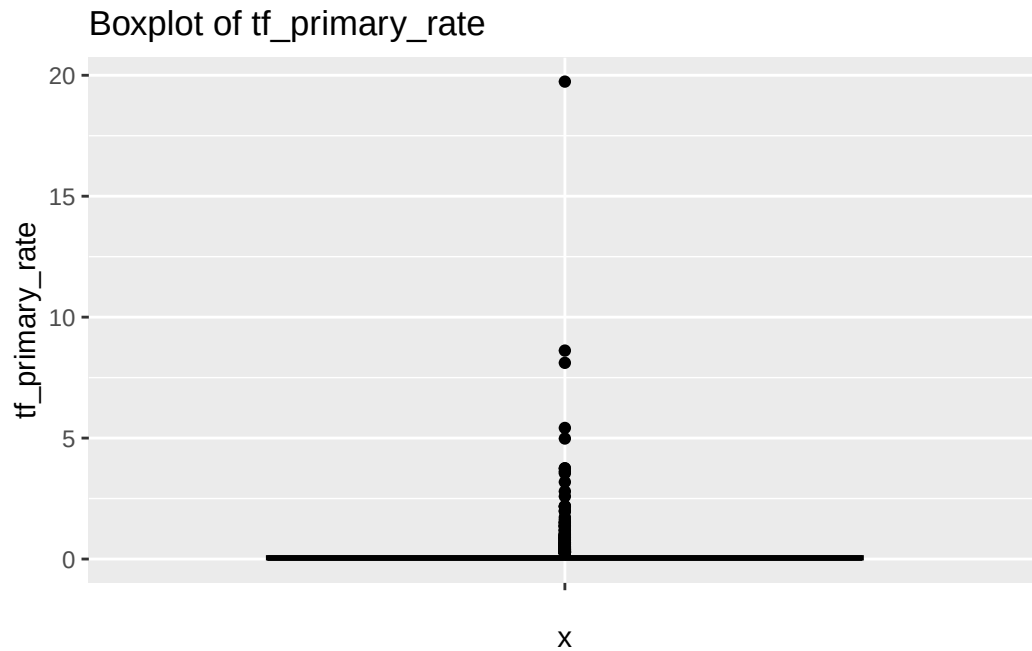
```
clean_data %>%  
  ggplot(aes(x = tf_primary_rate)) +  
  geom_histogram(binwidth = 0.1, fill = "green", color = "black", alpha = 0.7) +  
  labs(title = "Distribution of tf_primary_rate", x = "tf_primary_rate", y = "Frequency")
```



```
clean_data %>%  
  ggplot(aes(x = "", y = tf_blood_rate)) +  
  geom_boxplot(fill = "blue", color = "black") +  
  labs(title = "Boxplot of tf_blood_rate", y = "tf_blood_rate")
```



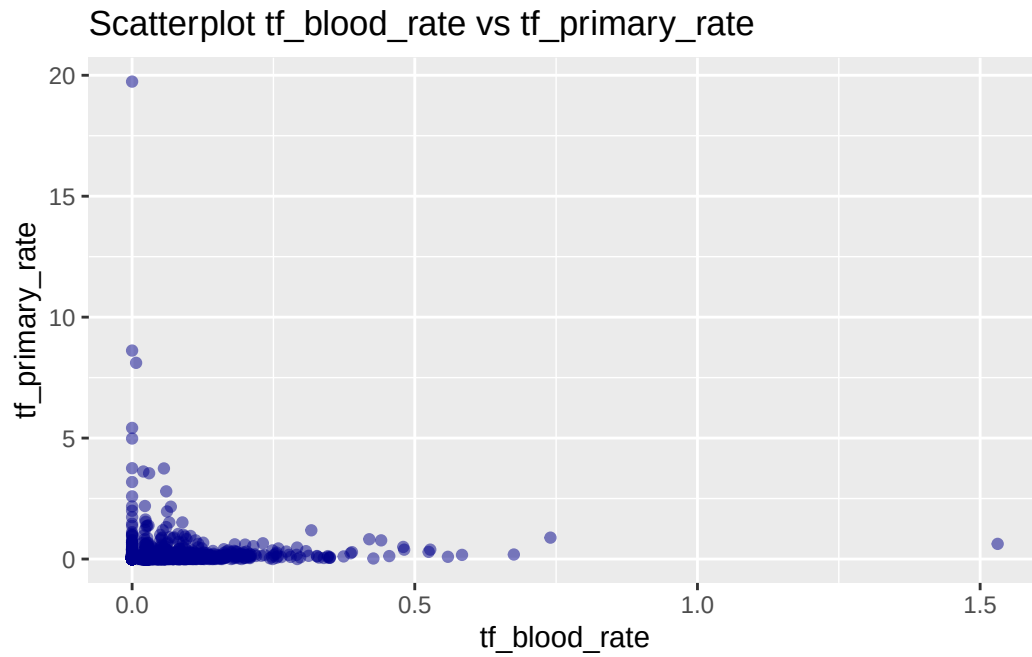
```
clean_data %>%  
  ggplot(aes(x = "", y = tf_primary_rate)) +  
  geom_boxplot(fill = "green", color = "black") +  
  labs(title = "Boxplot of tf_primary_rate", y = "tf_primary_rate")
```



Scatter plots

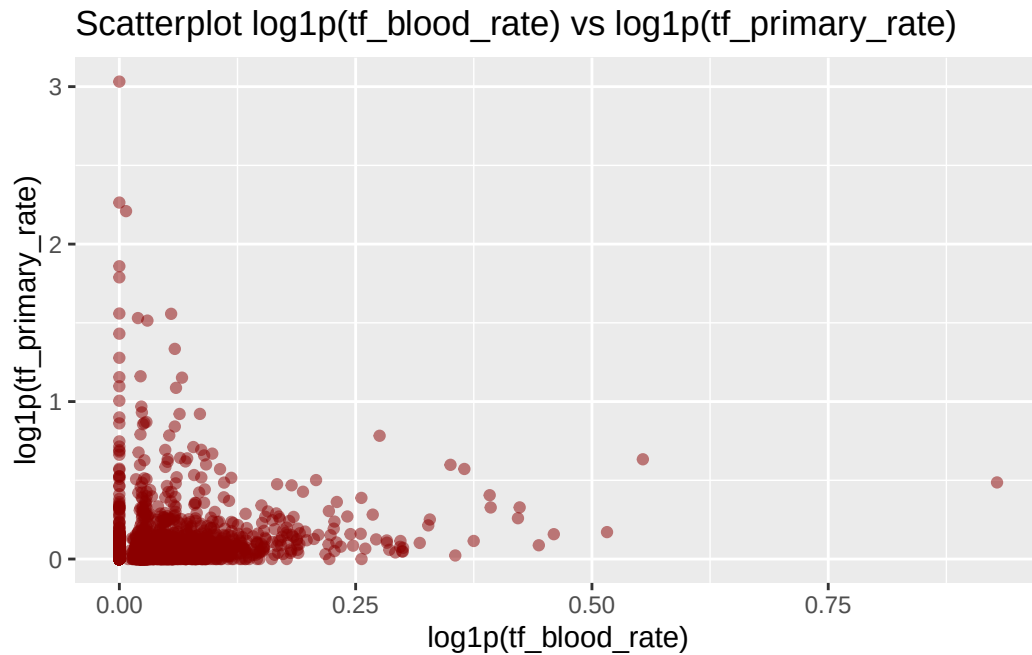
Scatterplot raw

```
clean_data %>%  
  ggplot(aes(tf_blood_rate, tf_primary_rate)) +  
  geom_point(alpha = 0.5, color = "darkblue") +  
  labs(title = "Scatterplot tf_blood_rate vs tf_primary_rate", x = "tf_blood_rate", y = "tf_
```



Scatterplot log1p transformed

```
clean_data %>%  
  ggplot(aes(log1p(tf_blood_rate), log1p(tf_primary_rate))) +  
  geom_point(alpha = 0.5, color = "darkred") +  
  labs(title = "Scatterplot log1p(tf_blood_rate) vs log1p(tf_primary_rate)", x = "log1p(tf_b
```

Distinction zero vs positive

```
clean_data <- clean_data %>%
  mutate(blood_pos = ifelse(tf_blood_rate > 0, 1, 0),
         primary_pos = ifelse(tf_primary_rate > 0, 1, 0))

table(clean_data$blood_pos, clean_data$primary_pos, dnn = c('blood_pos', 'primary_pos'))
```

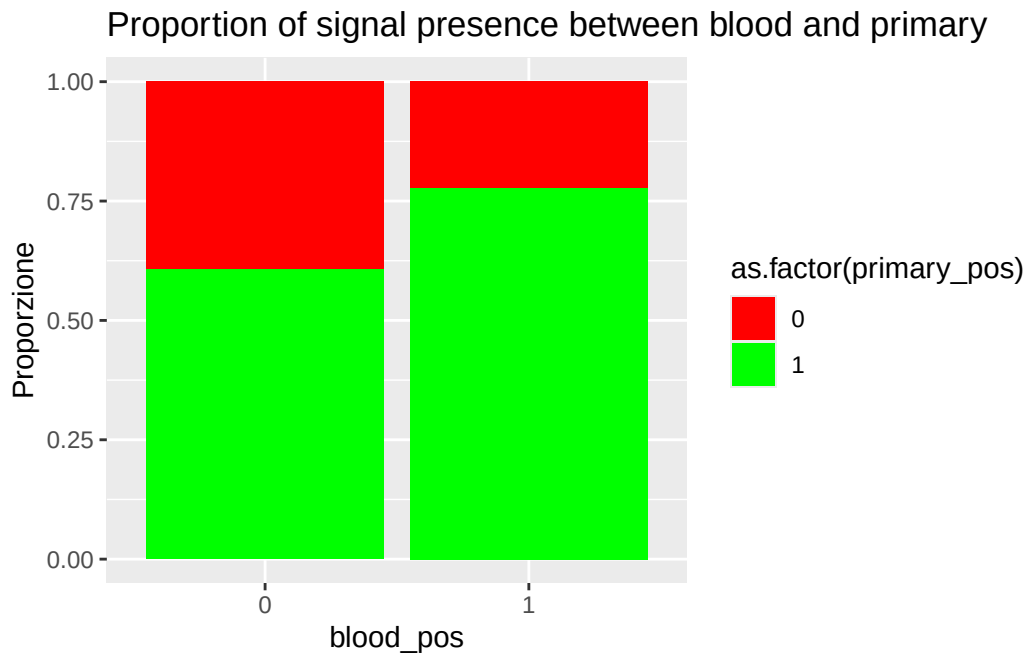
	primary_pos	
blood_pos	0	1
0	264	408
1	299	1043

```
chisq.test(table(clean_data$blood_pos, clean_data$primary_pos))
```

Pearson's Chi-squared test with Yates' continuity correction

```
data: table(clean_data$blood_pos, clean_data$primary_pos)
X-squared = 63.455, df = 1, p-value = 1.641e-15
```

```
ggplot(clean_data, aes(x = as.factor(blood_pos), fill = as.factor(primary_pos))) +
  geom_bar(position = "fill") +
  labs(title = "Proportion of signal presence between blood and primary", x = "blood_pos", y = "Proportion") +
  scale_fill_manual(values = c("red", "green"), labels = c("0", "1"))
```



Spearman correlation

```
positive_data <- clean_data %>%
  filter(blood_pos == 1 & primary_pos == 1)

cor.test(positive_data$tf_blood_rate[positive_data$blood_pos == 1],
  positive_data$tf_primary_rate[positive_data$primary_pos == 1],
  method = 'spearman')
```

Spearman's rank correlation rho

data: positive_data\$tf_blood_rate[positive_data\$blood_pos == 1] and positive_data\$tf_primary_rate[positive_data\$primary_pos == 1]
 S = 144483840, p-value = 1.401e-14

alternative hypothesis: true rho is not equal to 0
sample estimates:
rho
0.2359567

Relationship visualization

```
ggplot(positive_data, aes(x = log1p(tf_blood_rate), y = log1p(tf_primary_rate))) +  
  geom_point(alpha = 0.5, color = "darkred") +  
  geom_smooth(method = "loess", color = "blue") +  
  labs(title = "tf_blood_rate vs tf_primary_rate", x = "log1p(tf_blood_rate)", y = "log1p(tf_
```

